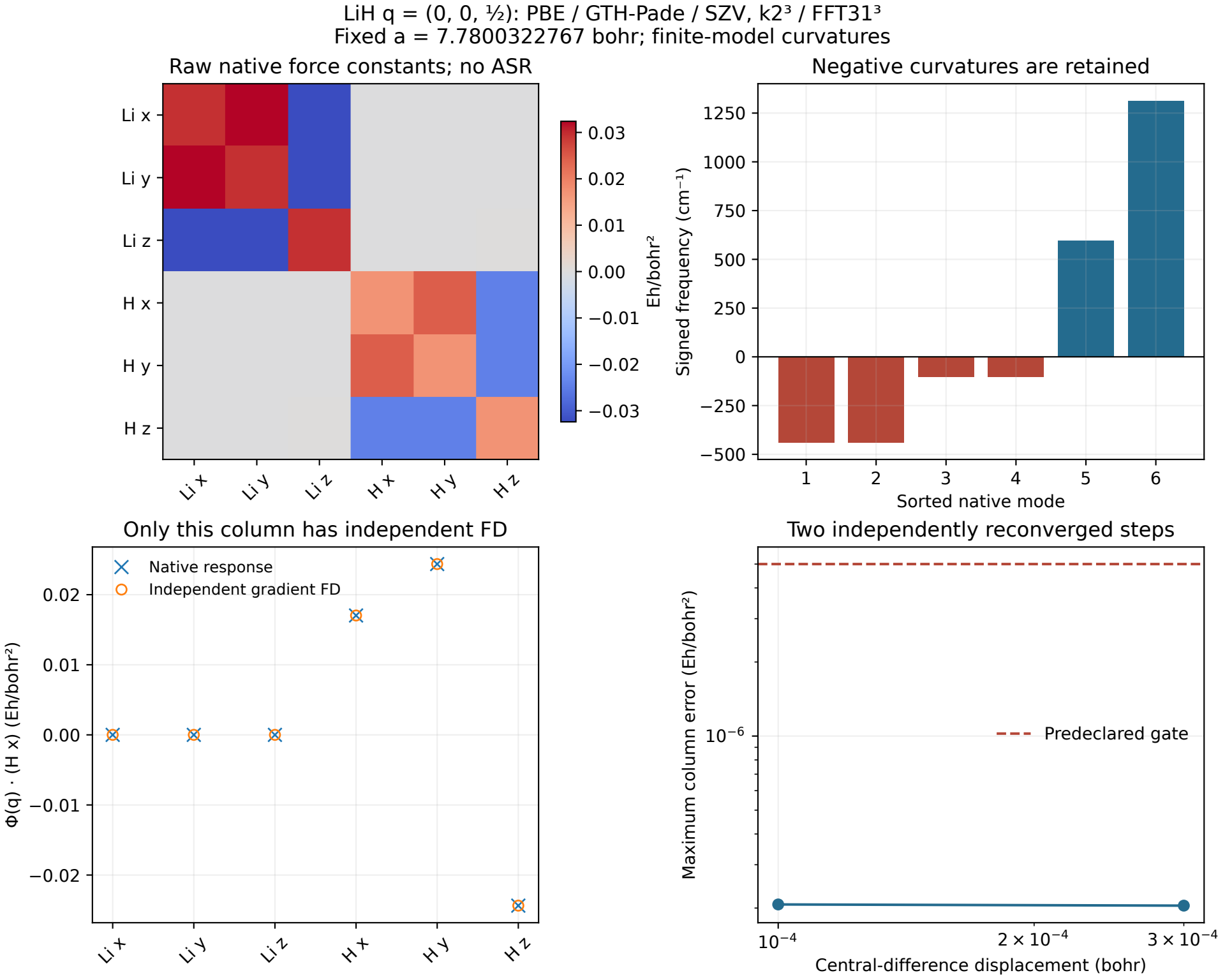
LiH q = (0, 0, ½): PBE / GTH-Pade / SZV, k2³ / FFT31³
Fixed a = 7.7800322767 bohr; finite-model curvatures
Raw native force constants; no ASR
Li x
Li y
Li z
H x
H y
H z
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
Eh/bohr²
Negative curvatures are retained
Signed frequency (cm⁻¹)
1250
1000
750
500
250
0
−250
−500
1
2
3
4
5
6
Sorted native mode
Only this column has independent FD
Native response
Independent gradient FD
Φ(q) · (H x) (Eh/bohr²)
0.02
0.01
0.00
−0.01
−0.02
Two independently reconverged steps
Maximum column error (Eh/bohr²)
Predeclared gate
10⁻⁶
10⁻⁴
2 × 10⁻⁴
3 × 10⁻⁴
Central-difference displacement (bohr)